Jewellery Shop Support Agent – Design Summary

1. Overall Flow

Vector Database Creation
1. Read jewellery product JSON.
2. Convert to structured JSON.
3. Split into chunks using Recursive Character Text Splitter.
4. Generate embeddings.
5. Store embeddings in a FAISS Vector Database.

Customer Support Agent
1. Receive customer query.
2. Agent processes the request.
3. Uses FAISS semantic search tool for product retrieval.
4. Retrieves relevant jewellery records.
5. Generates grounded natural-language responses.

2. Prompt Design

The prompt instructs the agent to behave as a jewellery shop assistant, answer only using retrieved catalogue information whenever possible, recommend suitable products, avoid hallucinating unavailable information, and respond politely in a conversational manner.

3. Memory Choice

The agent uses conversational memory configured to retain the last 100 chat messages. This enables multi-turn conversations, remembers customer preferences, supports follow-up questions, and improves shopping experience while keeping sufficient context without excessive memory usage.

4. Out-of-Stock Handling

If an item is unavailable, the agent politely informs the customer, does not fabricate availability, performs another semantic search for similar products, recommends alternatives based on category, metal, style, or price range, and asks whether the customer would like to view those alternatives.

5. Advantages

• Fast semantic search using FAISS.
• Reduced hallucinations through Retrieval-Augmented Generation (RAG).
• Easy catalogue updates by re-indexing.
• Modular Langflow architecture.
• Scalable for larger jewellery catalogues.